

Final Project  
Data Visualization and Data Driven Decision Making BSDVDDM1KU  
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# Danish Travel Habits

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# 1 Introduction

This project explores Danish travel behavior using multiple datasets that cover transportation trends, online consumer habits, and seasonal temperature patterns. The goal is to understand how different factors such as weather, travel type, and online activity influence when, where, and how Danes choose to travel.

Our focus is especially on seasonal behavior. Denmark experiences long, dark winters and mild summers, which we believe may strongly influence travel decisions. We are particularly interested in whether colder months drive increased international travel, and whether digital search trends reflect this desire to "escape" winter.

## 2 Hypothesis

Colder temperatures in Denmark are associated with increased interest in travel abroad, particularly by air travel as Danes seek to escape the winter season. We expect to see a rise in air travel during colder months, and a higher volume of travel related online search queries and purchases.

We based this hypothesis on the assumption that weather patterns influence both travel behavior and online interest. We also expected domestic travel (bus and trains) to show a different seasonal pattern, likely peaking during traditional holiday periods like summer and Christmas.

## 3 Data

This project includes 5 datasets about Danish travel habits. Here's what they cover:

- **CPH Airport Traffic:** This dataset tracks passenger traffic at Copenhagen Airport over time. It includes the destination country and the volume of passengers. Sadly there is a lot of missing data in this dataset so it needed some cleaning.
- **Average Temperatures in Denmark since 1901** This dataset is showing the average monthly temperatures across Denmark over more than a century. It can be used to explore climate trends and how weather conditions may influence travel behaviors, such as peak travel seasons or preferences.
- **Travel Type Data** This dataset provides monthly records of domestic travel trends in Denmark, starting from 2004. It includes three main categories: total travel activity, air travel, and travel by bus and rail.
- **Top Travel Related Search Queries** This dataset reflects the most common travel related search terms used by Danes online. It highlights interests and priorities, such as trending destinations or concerns. This dataset needed some translation so that it could be presented to a wider audience.
- **Online Research vs. Actual Purchase** This dataset compares the percentage of people who research products online and those who actually follow through with an online purchase across various product categories.

### 3.1 Dashboard #1

Link: [Tableau Dashboard #1 Link](#)

This dashboard gives a quick overview of Danish travel behavior between the chosen years (2004-2015 here), looking at:

- How people traveled (air vs. bus/rail vs. overall)

- Where they flew to the most
- How temperature varied across the year (to spot possible seasonal effects on travel)

### 3.2 Visualization #1 - Travel Type

For this visualization, we chose the grouped bar charts. They are perfect here, because they make it easy to compare travel modes side by side across months. You can spot how travel peaks in certain months and dips in others, reflecting seasonal fluctuations, such as higher travel in the summer and lower travel during late fall.

This bar chart style is ideal because it makes month to month comparisons obvious and it is easy to see each category side by side. Bars also help you quickly see which travel type is dominant in each month without having to decode a busy line chart or raw tables of numbers.

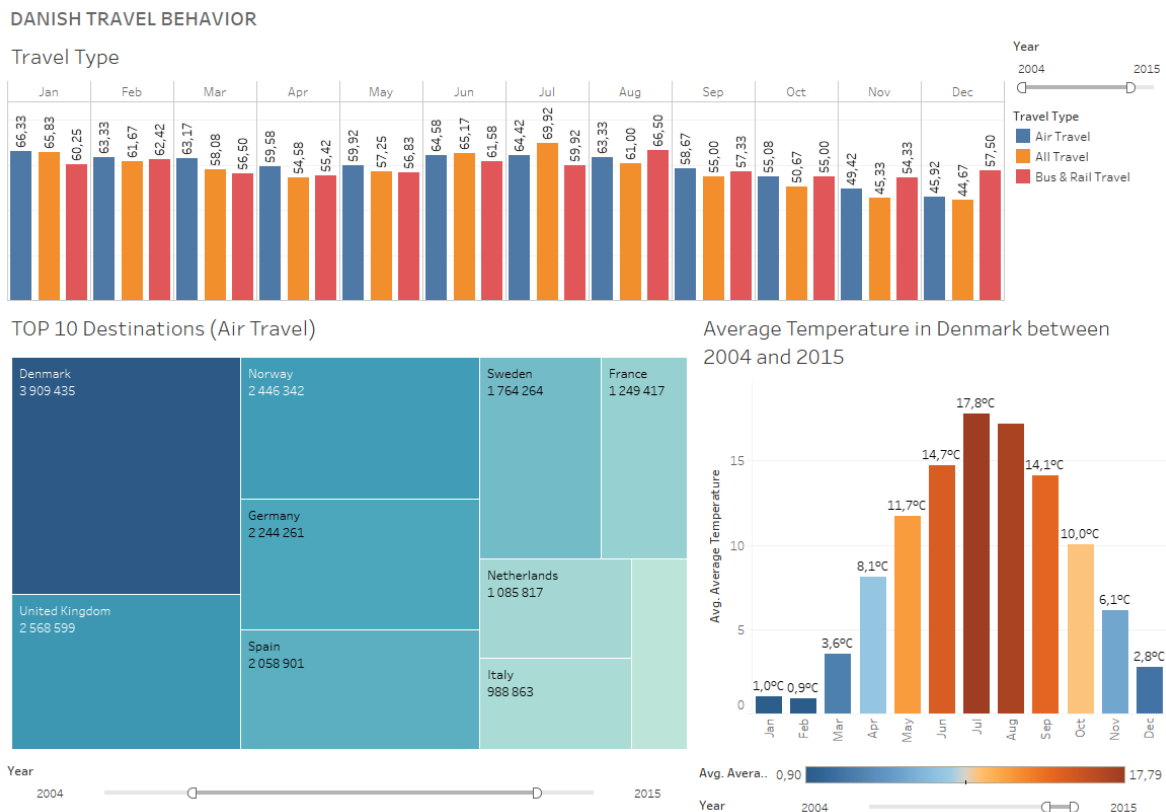


Figure 1: Dashboard #1

### 3.3 Visualization #2 - TOP 10 Destinations (Air Travel)

For this visualization, we chose the treemap, which shows the top 10 destinations (for air travel) ranked by total number of travelers. Each rectangle's size reflects that destination's volume, in comparison to others. Denmark appears to have the highest number of air travelers, with the United Kingdom and Norway following right after. The color shades and box sizes make it easy to visually compare destinations and see which ones dominate.

We chose a treemap because it is great for clearly showing the relative magnitude of multiple categories in a single view. It packs a lot of data into a small space, making the ranking instantly

obvious, and is more intuitive than a regular bar or line chart when there are many categories to compare.

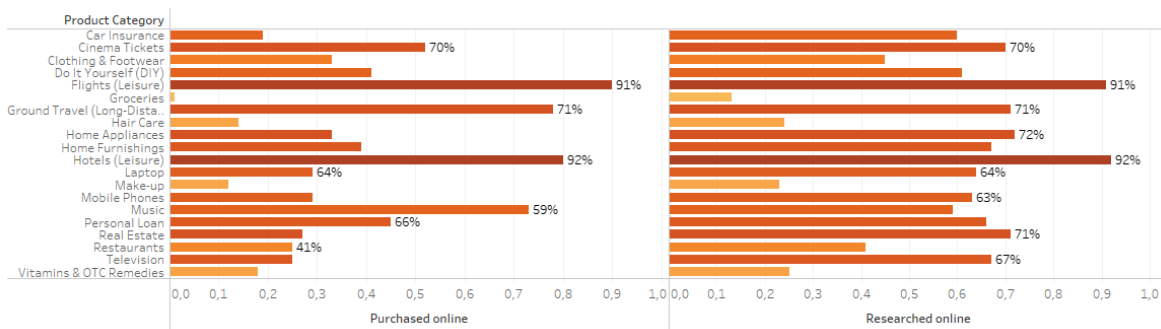
### 3.4 Visualization #3 - Average Temperature in Denmark

This chart shows how Denmark's average temperature changes throughout the year. The darker orange bars in summer emphasize the higher temperatures (around 17–18°C), while lighter blue bars in winter highlight the cold temperatures (around 0–2°C). This month by month bar chart is straightforward for comparing each month's average temperature and visually highlights the seasonal shifts by color coded ranges. It is clear which months are coldest and which are warmest.

The reason for choosing this type of visualization (color coded bar chart) is to make those seasonal fluctuations instantly recognizable. Each month stands on its own, and the color gradient (blue = cold, orange = warm) offers a quick visual of differences between months. You can also filter by year to see if the temperature profile changes over time.

#### DANISH TRAVEL AND PURCHASE HABITS

##### Research vs. Purchase



##### Most Popular Search Queries



Question  
☐ how  
☐ what  
☒ where  
☐ who  
☐ why

##### Average Temperature in Denmark between 2004 and 2015

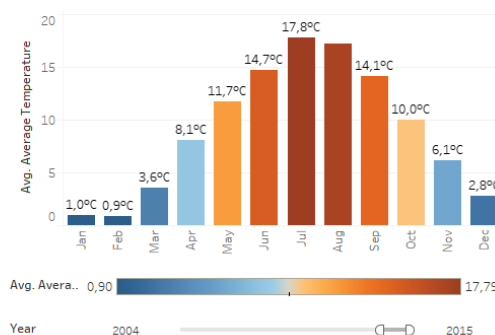


Figure 2: Dashboard #2

### 3.5 Dashboard #2

Link: [Tableau Dashboard #2 Link](#)

This dashboard focuses on online travel behavior, purchase habits and search queries related to travel in Denmark.

- What people research online vs. what they actually purchase
- Which travel related search queries are most common

- How temperature varied across the year

### 3.5.1 Visualization #1 - Research vs. Purchase

For this visualization, we chose the dual bar chart. It clearly contrasts the percentage of consumers who researched products online versus those who proceeded with an online purchase, organized by product category. The side by side bars make it easy to spot the differences between initial interest and final buying behavior, highlighting any significant gaps.

### 3.5.2 Visualization #2 - Most Popular Search Queries

For this visualization, we chose a treemap again, which showcases the most frequently searched travel queries, such as “Where to travel in April” or “Where to travel in January.”

We chose to color each tile differently so that the viewer can easily distinguish between queries. It is visually engaging and straightforward, especially when displaying short text snippets that need to stand out. Finally, there is a filter for the question word.

### 3.5.3 Visualization #3 - Average Temperature in Denmark

We have already covered this one. We put it there again to highlight when Danes choose to travel - mostly in the cold months to escape winter.

## 3.6 Conclusion

From the visualizations, we can conclude that:

- Danes tend to travel more in summer and aim to escape or reduce the cold during winter. Air travel peaks during the coldest months and in summer, pointing to both winter escape trips and summer holidays. Bus rail usage spikes in December, likely due to visits to family over the holidays. It’s also high in August, suggesting short domestic trips during the summer holidays.
- The UK, Norway, and other nearby countries are popular picks. The size of their boxes makes it clear how common quick trips to neighboring spots are. This supports the idea that short-haul, European city breaks are a regular thing.
- Searches spike around specific months (especially cold ones), suggesting that when it’s chilly, people either plan or dream about leaving.
- Many Danes do proper online research before traveling, and they actually follow through with booking.

## References

- [1] A. Cairo. *The Truthful Art: Data, Charts, and Maps for Communication*. Pearson Education, 2016.

[1]